Felipe Arthur Eckert

Software & Data Developer | Python · Data Pipelines · Machine Learning · Automation

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PROFESSIONAL SUMMARY

Electrical engineer from Poli-USP who moved from analyzing test data to building the tools that analyze it. In R&D at Hyundai Motor Brasil, I kept seeing valuable measurements stuck in manual spreadsheet work, so I started writing Python to automate it. That grew into an internal analysis app I designed and shipped end to end, from raw-file ingestion and validation to the statistical model and automated reports, which cut analysis time by about 80% and became a paper at SIMEA 2026. Since then I build and ship my own projects end to end, from a full-stack web app to an ML model.

SKILLS

Languages & Frameworks: Python, pandas, NumPy, SQL (PostgreSQL, SQLite), TypeScript, React, Next.js, Streamlit, C/C++

Data & ML: ETL pipelines, data cleaning and validation, statistics, scikit-learn, XGBoost, PyTorch, Optuna, SHAP

Engineering: Git/GitHub, GitHub Actions CI/CD, automated testing, REST APIs

AI Tools: AI-assisted development workflows (coding agents, custom skills and hooks), Microsoft 365 Copilot

Domain: Test, sensor and time-series data, embedded data acquisition, automotive R&D; quick to learn new domains

WORK EXPERIENCE

Hyundai Motor Brasil *Jan 2025 – Present*

Junior Research & Development Analyst – Energy Efficiency & Data Analytics

- Designed and shipped an internal coastdown analysis app (Python, Streamlit, SQLite): multi-source ingestion, weather-data sync, automated validation, statistical model and standardized reports; ~80% less analysis time.
- Kept the tool reliable as the team adopted it: 500+ automated tests covering the calculations and validation rules, and a performance review that sped up the automatic selection of valid runs.
- Co-authored the first paper published by Hyundai Motor Brasil's R&D center (SIMEA 2026), presenting the data-driven coastdown method behind the tool.
- Won 1st place at the Hyundai Prompt-a-thon (Sep 2026) with R&D colleagues: a Microsoft 365 Copilot hub that consolidates multi-source information into executive reports.

Poli-USP BAJA SAE Team *Mar 2018 – Dec 2024*

Powertrain Director (2021 – 2024) · Powertrain Coordinator (2018 – 2021)

- Co-developed a custom telemetry data logger with the electronics sub-team, expanding sensor channels and enabling data-driven vehicle setup decisions across competition events.
- Led development of the team's first 4x4 gearbox from concept to competition, coordinating suppliers on cost, lead time, and manufacturability under tight deadlines.

SELECTED PROJECTS

- Paper trading simulator (live demo): full-stack Next.js and PostgreSQL app with auth, a REST API (API keys, rate limits, idempotent orders), a backtester and ~220 tests in CI; deployed on Vercel.
- Vehicle energy efficiency estimator (GitHub): ETL over Brazil's PBEV 2026 label data, then XGBoost vs. random forest and PyTorch networks under grouped nested CV with Optuna and monotonic constraints (5.8% error); Streamlit app.
- Finale (GitHub): one-knob VST3 mastering plugin in C++ (JUCE), built with a friend: real-time audio engine, loudness metering, ~180 tests and CI.

EDUCATION

Escola Politécnica da USP (Poli-USP) *2018 – 2025*

B.S. in Electrical Engineering — Emphasis: Automation & Control | Specialization: Automotive Engineering

LANGUAGES

Portuguese: Native **English:** Advanced **Spanish:** Conversational

PUBLICATIONS & AWARDS

- Co-author, “Data-Driven Coastdown Analysis”, SIMEA 2026 (Blucher Engineering Proceedings).
- 1st Place Overall & 2nd Place (Logic) — Brazilian Knowledge Olympiad (OBS)